ProbeRoute: Probes as Routing Priors for Frozen-Backbone Multi-Token Prediction

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Abstract

Frozen autoregressive language models are trained for next-token prediction, but their hidden states can still encode information about several future tokens. ProbeRoute tests whether that latent structure can be converted into a better explicit multi-token adapter without unfreezing the backbone. The method first trains horizon-specific future-token probes across depth, then uses probe-derived top-5 scores to initialize a sparse top-m router over frozen hidden states. Under a stage-gated protocol with mandatory probes, screening baselines, a finalist-selection step, and a final 1B rerun, the resulting sparse adapter beats the strongest selected dense frozen-backbone baseline on the paper's two headline held-out metrics. On the final 1B comparison, ProbeRoute improves `mean_top1_h2_h4` from 0.1162 to 0.1172 and the acceptance proxy `mean_accept_len` from 1.1110 to 1.1188, while changing `mean_nll_h1_h4` only from 5.3769 to 5.3780. The mechanistic story is equally important. Probe heatmaps reveal horizon-dependent depth structure at both 410M and 1B, the learned sparse router places mass on the same layer bands, and the random-initialization ablation is weaker than probe-initialized sparse routing. By contrast, loss warmup and deeper far-horizon heads do not improve over the base sparse configuration at the tested budget. The evidence therefore supports a focused conclusion: in this frozen-backbone setting, future-token probes are not merely descriptive diagnostics; they are a useful routing prior for explicit multi-token prediction.

1 Introduction

Explicit multi-token prediction is attractive because it asks a language model for more than the next token at each decoding position. In principle, that creates room for richer future supervision during adaptation and for block-level acceptance proxies at evaluation time. In practice, however, a pretrained autoregressive backbone is optimized for one-step prediction, so a naive multi-token adapter can easily read from the wrong part of the network. The central question in this paper is whether a frozen backbone already contains enough horizon-specific future structure to support a better adapter than standard last-layer or dense layer-mixing baselines.

ProbeRoute answers that question with a deliberately constrained recipe. We keep the pretrained backbone frozen, probe every layer for future-token predictability, and use the probe scores to initialize a sparse per-horizon layer router. The key idea is simple: if future-token information is distributed non-uniformly across depth, then probes should identify which layer bands matter for each horizon, and that information should be useful as an adaptation prior rather than remaining a diagnostic artifact.

The resulting study is intentionally focused. It does not ask whether sparse routing universally dominates dense routing, whether frozen adapters beat full finetuning, or whether the acceptance proxy is a direct throughput benchmark. Instead, it asks a sharper question that the artifact packet can answer cleanly: *does probe-initialized sparse routing improve frozen-backbone multi-token prediction relative to the strongest dense baseline selected under the same screening and final-rerun protocol?*

The completed evidence supports a positive answer. Mandatory probe runs at 410M and 1B reveal non-flat, horizon-dependent depth structure. Screening then identifies a strong dense weighted-hidden-state baseline and reruns it at the final budget as the finalist comparator. Against that selected dense finalist, the final 1B sparse run improves the future-token top-1 aggregate and the acceptance proxy while changing aggregate NLL only minimally. The ablation stack narrows the interpretation further: random initialization weakens the sparse model, while extra warmup and deeper far-horizon heads do not explain the gain.

This yields three contributions that are fully supported by the packet bundled with this manuscript:

- We show that future-token probe heatmaps reveal horizon-dependent depth structure at both 410M and 1B, with shorter horizons peaking later and farther horizons shifting earlier in depth.
- We turn those probe measurements into a concrete frozen-backbone adapter—a sparse top-m layer router initialized from probe top-5 scores—and evaluate it against screening-selected last-layer and dense weighted-hidden-state baselines.
- We find that the final 1B sparse run beats the strongest selected dense baseline on held-out future-token top-1 and acceptance metrics, and that the random-init ablation is the one that meaningfully weakens the sparse recipe.

2 Related Work and Positioning

This paper touches three nearby lines of work, but it occupies a narrower space than any of them.

Multi-token prediction and multi-head decoding. Recent work has revisited the standard next-token objective by asking language models to predict multiple future tokens in parallel. Gloeckle et al. [4] study multi-token prediction during large-scale language-model training and show gains in sample efficiency and downstream generation quality. A separate line of work uses multiple decoding heads primarily for faster inference. Speculative decoding introduces a draft-and-verify framework for exact acceleration without changing the output distribution [8]. Medusa replaces a separate draft model with lightweight decoding heads on top of the base model [3], while Hydra improves those heads by making them sequentially dependent [2]. EAGLE pushes the same acceleration theme further with feature-level extrapolation for speculative sampling [10]. Our setting is different from all of these in two ways: the backbone remains frozen throughout, and the main question is not raw decoding speed but whether horizon-specific future structure can be exploited to build a stronger frozen-backbone multi-token adapter.

Probes as analysis tools. The probe stage draws on a well-established interpretability tradition. Linear classifier probes were introduced as a way to inspect intermediate representations without perturbing the original training dynamics [1]. Later work emphasized that probe accuracy alone can be misleading unless probe capacity and selectivity are controlled carefully [5]. We adopt the core probing intuition from that literature, but use probes differently. In this paper, probes are not only descriptive diagnostics for analyzing hidden states after the fact. They are used operationally to define a routing prior for the downstream adapter.

Frozen-backbone adaptation. The broader motivation also overlaps with parameter-efficient adaptation methods, which keep most or all pretrained parameters fixed while learning a lightweight task-specific interface. Adapters [6], prefix tuning [9], and LoRA [7] all pursue that general objective. ProbeRoute differs in what is being adapted. Instead of inserting low-rank weight updates or learned prompts, the method learns how to access existing hidden-state depth structure in a frozen model, with probes supplying the initial guess for which layers matter at each horizon.

Taken together, these neighboring literatures help clarify the paper's contribution. We are not proposing a general speculative decoding system, a new pretraining recipe for large language models, or a generic parameter-efficient fine-tuning method. The contribution supported by the packet is narrower and more mechanistic: future-token probes reveal usable depth structure, and that structure can be converted into a sparse routing prior that improves frozen-backbone multi-token prediction relative to the strongest dense baseline selected under the same protocol.

3 Method

3.1 Problem setup

Let a frozen autoregressive backbone produce hidden states $h_t^{(\ell)}$ at token position t and layer $\ell \in \{0, \ldots, L-1\}$. The adaptation problem is to predict future tokens x_{t+k} for horizons $h \in \{1, 2, 3, 4\}$ while changing only a lightweight router and horizon heads. The study compares three adapter families:

- **Last-layer baselines**: decode each horizon only from the final hidden state.
- **Dense weighted-hidden-state (WHS) baselines**: learn a dense mixture over all layers for each horizon.
- **Sparse top-m routing**: learn a mixture over only a small horizon-specific subset of layers.

The proposed method differs from the sparse random-init ablation only in how that subset and its initial weights are chosen.

3.2 Probe stage

The first stage trains horizon-specific future-token probes over every layer of the frozen backbone. The packet's canonical probe defaults use rank-64 probes, a frozen backbone, and validation top-5 as the initialization metric. For each layer ℓ and horizon k, the probe stage produces a validation score $q_{\ell,k}$, which we treat as a measurement of how useful that layer is for predicting x_{t+k}.

The probes are not an end in themselves. Their role is to summarize a layer × horizon landscape before the main adapter is trained. In the final paper narrative, this is the key methodological shift: probe heatmaps are used as an *input* to the adapter, not merely as a post hoc visualization.

3.3 From probe scores to sparse routers

For dense WHS, each horizon learns a full mixture over all layers. For ProbeRoute, we instead form a sparse support set S_k by taking the top-m layers under probe-derived top-5 scores, with $m = 4$ in all main sparse runs. The router is then initialized from z-scored probe top-5 values on that support and trained jointly with the horizon heads while the backbone remains frozen.

Writing the learned router weights as $\alpha_{\ell,k}$, the mixed representation for horizon k is

$$\tilde{h}_t^{(k)} = \sum_{\ell \in S_k} \alpha_{\ell,k} \, \mathrm{LN}\left(h_t^{(\ell)}\right), \tag{1}$$

where $\mathrm{LN}(\cdot)$ is the stateless layer normalization used by the packet's canonical MTP adapter. A residual MLP head then maps $\tilde{h}_t^{(k)}$ to logits for the token at horizon k.

The dense WHS baselines use the same frozen-backbone setting and the same family of horizon heads. The comparison therefore isolates the routing strategy rather than head capacity or backbone updates.

4 Experimental Protocol

4.1 Data, models, and budgets

The packet's experiment plan uses deterministic processed FineWeb-Edu subsets throughout. Probes are run at 410M

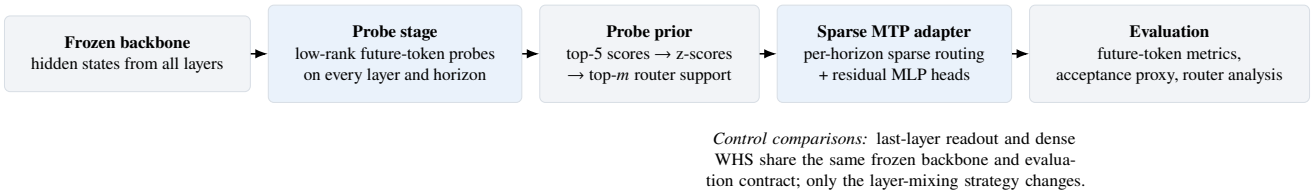

Figure 1: Study design. ProbeRoute converts probe measurements into a routing prior. Every stage keeps the pretrained backbone frozen; the comparison is therefore about how future information is accessed, not about unfreezing or large-scale retraining.

and 1B with sequence length 1024 and budgets of 5M train / 1M validation / 1M test tokens. Screening and ablation runs use Pythia-1B at sequence length 2048 with 20M / 2M / 2M token budgets. Final 1B reruns use 50M / 5M / 5M token budgets. The low-cost confirmatory path reruns the ProbeRoute recipe at 410M with a second seed and a 20M / 2M / 2M budget.

The canonical MTP defaults in the plan are fixed across the relevant runs: frozen backbones, horizons $\{1, 2, 3, 4\}$, residual-MLP horizon heads, stateless layer normalization, router temperature 0.5, entropy penalty $\beta = 10^{-3}$, and top-$m = 4$ for sparse routing.

4.2 Screening and finalist selection

The five mandatory screening runs are:

(i) `BASE_LAST_LINEAR_1B`,
(ii) `BASE_LAST_MLP_1B`,
(iii) `BASE_DENSE_WHS_RANDOM_1B`,
(iv) `BASE_DENSE_WHS_PROBE_1B`, and
(v) `MAIN_SPARSE_PROBE_1B`.

Finalist selection is deliberately conservative: only completed *non-sparse* screening baselines are eligible, and the primary score is the validation future-token top-1 aggregate at the selected best checkpoint. Under that rule, Dense WHS Probe-Init is the finalist baseline and is promoted to the 50M-token final comparison against the final sparse rerun.

4.3 Metrics

The study tracks future-token quality at horizons 1–4 using top-1, top-5, and negative log-likelihood (NLL). The paper's main aggregate metric is the mean top-1 over horizons 2–4, because it emphasizes the explicitly multi-token part of the task rather than allowing the horizon-1 term to dominate. The second headline metric is mean accepted-prefix length, measured with $K_{\max} = 4$ on deterministic prefixes. Following the experiment plan, the acceptance proxy rolls the frozen base model greedily for up to four tokens, compares the proposed MTP block against that rollout, records the longest matching prefix, appends exactly one base-greedy token, and repeats. This is a practical proxy, not an end-to-end throughput benchmark.

All non-probe scientific results are evaluated from best checkpoints chosen on validation data. That contract is part of the evidence policy bundled with the packet and is important to the paper's interpretation.

5 Probes Reveal Depth Structure Across Horizons

The probe results establish the paper's mechanism story before the main adapter is trained. At 1B, the best layer for horizon 1 is layer 13, while the best layers shift earlier for horizons 2–4 (layers 10, 9, and 9 respectively). At 410M, the shift is even larger: the best layer moves from 22 at horizon 1 to 15, 13, and 12 for horizons 2–4. The heatmaps are therefore not flat; they show that the depth band with the strongest future-token signal depends on how far into the future the model is asked to predict.

The spread of probe top-5 accuracy across layers reinforces the same point. At 1B, the range from worst to best layer is 5.90pp for horizon 1, 3.11pp for horizon 2, 1.59pp for horizon 3, and 1.12pp for horizon 4. At 410M, the corresponding spreads are 7.07pp, 3.35pp, 1.18pp, and 0.78pp. The far-horizon signal is weaker, but it is still structured rather than uniform.

The router analysis in figure 2 shows that the learned sparse adapter respects this measured structure. In the final 1B sparse run, the selected layer sets are tightly concentrated around the best probe layers for each horizon. Horizon 1 remains late, horizons 3 and 4 move earlier, and horizon 2 falls between them. This alignment matters because it links the diagnostic probe stage to the downstream adapter rather than treating the two as unrelated analyses.

Together, the mandatory probes support the paper's first claim: future-token information in a frozen autoregressive backbone is depth-structured and horizon-dependent, and that structure is strong enough to guide an explicit routing prior.

6 Final 1B Comparison

6.1 Screening selects a strong dense baseline

Before making any headline claim, the study runs a full screening sweep at the 20M-token budget. This step is important because it prevents the final comparison from using a weak straw baseline. Among the non-sparse screening baselines, dense weighted-hidden-state mixing with probe initialization achieves the highest validation `mean_top1_h2_h4` and is therefore promoted to the final-budget rerun.

The screening results already suggest a useful distinction between adaptation goals. The last-layer residual-MLP baseline attains the best screening NLL, but it lags badly on `mean_top1_h2_h4`. By contrast, the dense probe-initialized WHS baseline and the sparse screening run are much closer on the explicitly multi-token metrics that the paper cares

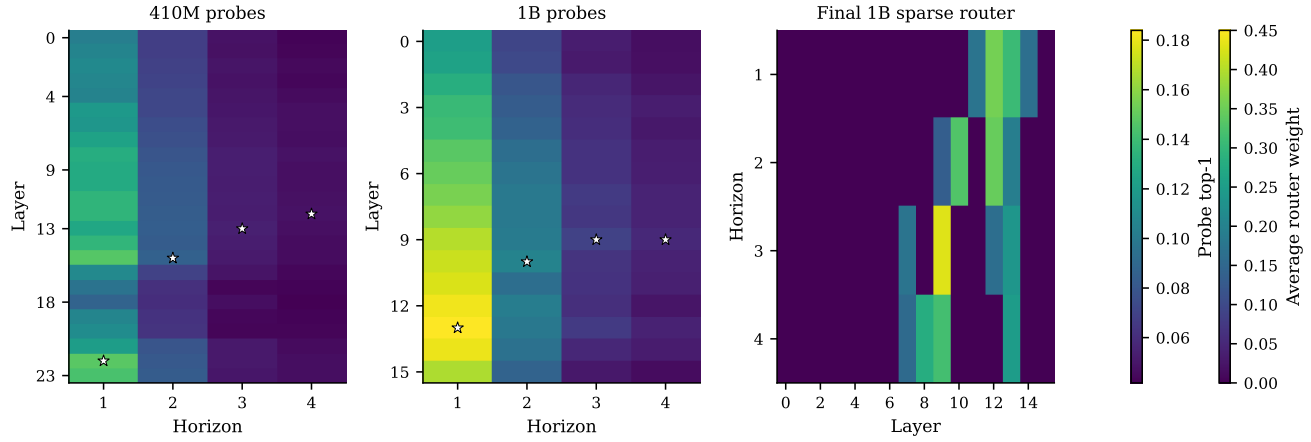

Figure 2: Mandatory probe heatmaps and final sparse-router support. Left and center: probe top-1 heatmaps for the 410M and 1B mandatory probe runs. The brightest layers move earlier as the prediction horizon increases. Right: average learned router weights for the final 1B sparse model. The sparse adapter concentrates its mass in the same late-to-mid depth bands identified by the 1B probes. In particular, the final sparse router selects layers {12,13,11,14}, {12,10,13,9}, {9,13,7,12}, and {9,8,13,7} for horizons 1–4, closely matching the probe peaks at layers 13, 10, 9, and 9.

Table 1: Screening results at the 20M-token budget. The finalist baseline is selected from non-sparse baselines by validation `mean_top1_h2_h4`. The sparse screening run is shown for context but is not eligible for baseline selection. Higher is better for `mean_top1_h2_h4` and `mean_accept_len`; lower is better for `mean_nll_h1_h4`.

Method	Init	Val top-1 h2–4	Test top-1 h2–4	Test accept len	Test NLL h1–4
Last-layer linear	–	0.0932	0.0941	0.6530	6.1853
Last-layer residual MLP	–	0.1004	0.1011	1.0712	**5.4016**
Dense WHS	random	0.1031	0.1039	0.8690	5.7361
Dense WHS	probe-init	**0.1056**	**0.1066**	0.9987	5.6230
Sparse top-m (proposed)	probe-init	0.1053	0.1061	0.9863	5.6412

Dense WHS Probe-Init is the strongest non-sparse baseline by validation `mean_top1_h2_h4` and becomes the final comparator. The sparse screening run trails narrowly at screening budget, which makes the later final-budget win more informative.

Table 2: Final 1B comparison at the 50M-token budget. Both finalists use probe-derived initialization. ProbeRoute wins on the two headline held-out metrics while changing mean NLL only minimally.

Model	Test top-1 h2–4	Test accept len	Test NLL h1–4
Dense finalist	0.1162	1.1110	**5.3769**
ProbeRoute (proposed)	**0.1172**	**1.1188**	5.3780
Δ (sparse−dense)	+0.00099	+0.00781	+0.00110

The NLL change is only +0.00110 in absolute terms, which is about 0.02% relative to the dense finalist.

about. This is why the final comparison should be read as a comparison between two serious frozen-backbone MTP adapters, not between a proposed method and a deliberately weak baseline.

6.2 ProbeRoute wins the final held-out comparison

The final 1B comparison is the paper's central quantitative result. Relative to the selected dense finalist, ProbeRoute improves the future-token top-1 aggregate from 0.1162 to 0.1172 and mean accepted-prefix length from 1.1110 to 1.1188. The dense finalist keeps a tiny likelihood advantage in the aggregate, but the difference in mean NLL is only 0.00110. In relative terms, ProbeRoute improves the main multi-token accuracy aggregate by roughly 0.85% and the acceptance proxy by roughly 0.70%, while the NLL change is only about 0.02%.

The per-horizon view in figure 3 is especially useful. Top-1 improves at every horizon, not just in the aggregate. The largest gain appears at horizon 3, which is where explicit multi-token structure is already far enough from the next-token objective to make routing matter, but not so far that the signal collapses. The NLL trade-off is concentrated at horizon 4; the sparse model is actually slightly better on NLL at horizons 1–3.

A second, subtler point is that the final 50M-token rerun reverses the narrow screening deficit of the sparse model relative to the dense finalist. We do not turn that into a scaling claim, because the packet contains only one final-budget comparison, but the reversal does make the final win more credible: the result is not inherited from screening, it emerges under the deliberate final-rerun protocol.

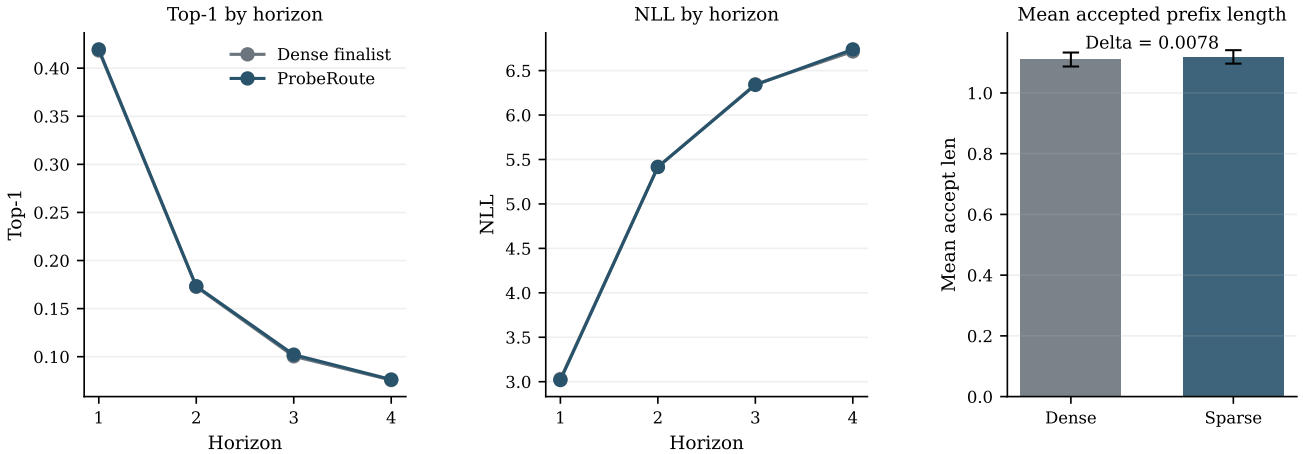

Figure 3: Final 1B metric profile. Left: the sparse final run improves top-1 at every horizon. Center: NLL is slightly better for the sparse model at horizons 1–3 and slightly worse at horizon 4, which explains the near-neutral aggregate likelihood change. Right: the sparse model also improves the acceptance proxy `mean_accept_len` on the final test set.

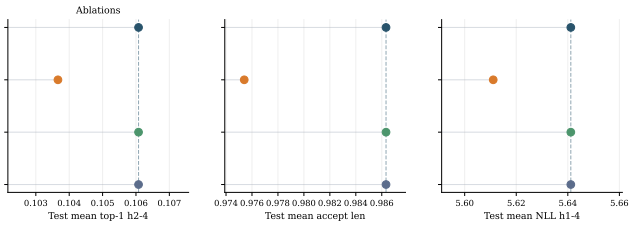

Figure 4: Screening-budget sparse ablations. The base sparse configuration uses probe initialization. Random initialization weakens both `mean_top1_h2_h4` and `mean_accept_len`. The warmup and deephead variants coincide with the base sparse metrics at reported precision. Lower NLL does not rescue the random-init ablation because the study's targeted future-token and acceptance metrics both drop.

7 Ablations, Confirmatory Evidence, and Discussion

7.1 The probe prior is the decisive ingredient

The sparse ablations are informative precisely because they do *not* all move in different directions. The random-init ablation is the one that clearly weakens the method on the targeted metrics, while warmup and deeper far-horizon heads do not improve on the base sparse configuration at the packet's reported precision.

At the 20M-token budget, the base sparse screening run reaches a future-token top-1 aggregate of 0.1061 and a mean accepted-prefix length of 0.9863. Removing the probe prior drops these to 0.1037 and 0.9754. That is the cleanest mechanism test in the paper: changing the sparse model's initialization while leaving the rest of the frozen-backbone recipe intact weakens the two metrics that define success in this study.

The warmup and deephead ablations tell a different story. Both match the base sparse configuration at the reported aggregate precision. This makes the positive result more convincing, not less: the final gain is not well explained by generic optimization polish, loss-scheduling tricks, or extra head complexity. The paper's most defensible interpretation is therefore also its sharpest one—the probe prior itself is the decisive ingredient.

7.2 Confirmatory fallback

The project's low-cost confirmatory path reruns ProbeRoute at 410M with a second seed rather than using the more expensive 2.8B extension. The confirmatory run completes with a future-token top-1 aggregate of 0.0989, a mean accepted-prefix length of 0.9852, and an aggregate NLL of 5.7844. This is supporting evidence only. It does not replace the final 1B headline comparison, and it is not used to claim cross-scale superiority. Its value is narrower: it shows that the same recipe completes cleanly in a second-seed, smaller-model setting and preserves the qualitative shape of a functioning ProbeRoute adapter.

7.3 Discussion and limitations

The strongest supported story in the packet is now clear. Future-token probes reveal useful depth structure in frozen autoregressive backbones. Turning those probes into a sparse routing prior yields a better explicit multi-token adapter than the strongest selected dense frozen-backbone baseline under the same screening and final-rerun contract. The ablations then isolate why that happens: probe initialization matters, while the tested warmup and deeper-head variants do not explain the gain.

The study remains deliberately narrow. It is confined to the Pythia family, frozen-backbone adaptation, a single final 1B seed, and a low-cost 410M confirmatory fallback rather than a larger confirmatory extension. The effect sizes are real but modest, and `mean_accept_len` remains a block-acceptance proxy rather than a production-throughput measurement. We therefore present the result as a focused contribution paper, not as a universal claim about all multi-token predictors or all sparse adapters.

8 Conclusion

This paper asked a targeted question: can future-token probes be turned into a useful routing prior for explicit multi-token prediction on top of a frozen pretrained backbone? Under the completed ProbeRoute protocol, the answer is yes. Mandatory probe runs at 410M and 1B show horizon-dependent depth structure, the final sparse router places mass on the same layer bands, the final 1B sparse model beats the strongest selected dense baseline on held-out `mean_top1_h2_h4` and `mean_accept_len`, and the random-init ablation is the one that meaningfully weakens the sparse recipe.

That combination of results is what makes the contribution coherent. The paper does not rely on a single positive comparison or on a process-heavy narrative. It makes a specific mechanistic claim and backs it with probes, screening, final reruns, and ablations. Within that frozen-backbone setting, probes are not only descriptive. They are useful routing priors.

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A Additional Tables and Provenance

A.1 Probe summary

Table 3: Best probe layer by horizon and top-5 spread across layers. The best layer shifts earlier as the horizon increases in both mandatory probe runs. Top-5 spread is reported in percentage points.

Model	Horizon	Best layer	Best top-1 (%)	Top-5 spread (pp)
1B	1	13	18.39	5.90
1B	2	10	10.54	3.11
1B	3	9	6.97	1.59
1B	4	9	5.82	1.12
410M	1	22	14.76	7.07
410M	2	15	8.57	3.35
410M	3	13	5.41	1.18
410M	4	12	4.86	0.78

A.2 Ablation table

Table 4: Aggregate test metrics for the sparse screening configuration and its planned ablations.

Variant	Test top-1 h2–4	Test accept len	Test NLL h1–4
ProbeRoute	0.1061	0.9863	5.6412
Sparse random-init	0.1037	0.9754	5.6111
Sparse + warmup	0.1061	0.9863	5.6412
Sparse + deephead	0.1061	0.9863	5.6412

A.3 Resource summary

Table 5: Execution summary for the authoritative runs included in the paper packet.

Experiment	Stage	Backbone	Train budget	Best step
PROBE 410M	probe	Pythia-410m	5M	77
PROBE 1B	probe	Pythia-1b	5M	77
BASE DENSE WHS PROBE 1B	screen	Pythia-1b	20M	153
MAIN SPARSE PROBE 1B	screen	Pythia-1b	20M	153
FINAL BEST BASELINE 1B	final	Pythia-1b	50M	382
MAIN SPARSE PROBE 1B FINAL	final	Pythia-1b	50M	375
ABL SPARSE RANDOM 1B	ablation	Pythia-1b	20M	153
ABL SPARSE WARMUP 1B	ablation	Pythia-1b	20M	153
ABL SPARSE DEEPHEAD 1B	ablation	Pythia-1b	20M	153
CONFIRM BEST 410M SEED2	confirm	Pythia-410m	20M	153

A.4 Resolved config summary

Table 6: Resolved experiment identities from the packet's config summary table.

Experiment	Stage	Backbone	Layer mix	Router init	Seq. length
ABL SPARSE DEEPHEAD 1B	ablation	pythia-1b	sparse-topm	probe-zscore-top5	2048
ABL SPARSE RANDOM 1B	ablation	pythia-1b	sparse-topm	random	2048
ABL SPARSE WARMUP 1B	ablation	pythia-1b	sparse-topm	probe-zscore-top5	2048
CONFIRM BEST 410M SEED2	confirm	pythia-410m	sparse-topm	probe-zscore-top5	2048
FINAL BEST BASELINE 1B	final	pythia-1b	dense-whs	probe-zscore-top5	2048
MAIN SPARSE PROBE 1B FINAL	final	pythia-1b	sparse-topm	probe-zscore-top5	2048
PROBE 1B	probe	pythia-1b	sparse-topm	none	1024
PROBE 410M	probe	pythia-410m	sparse-topm	none	1024
BASE DENSE WHS PROBE 1B	screen	pythia-1b	dense-whs	probe-zscore-top5	2048
BASE DENSE WHS RANDOM 1B	screen	pythia-1b	dense-whs	random	2048
BASE LAST LINEAR 1B	screen	pythia-1b	last-layer	none	2048
BASE LAST MLP 1B	screen	pythia-1b	last-layer	none	2048
MAIN SPARSE PROBE 1B	screen	pythia-1b	sparse-topm	probe-zscore-top5	2048

A.5 Claim-evidence map

Table 7: Artifact-backed claim map bundled with the paper packet.

Claim	Evidence source	Status
C1	Probe score matrices from `PROBE_1B` and `PROBE_410M`	supported
C2	Final 1B comparison registry (`main_results.csv`)	supported
C3	Final acceptance-proxy metrics for the dense and sparse 1B finalists	supported
C4	Screening-budget sparse ablation registry (`ablation_results.csv`)	supported